

$$P \vee (Q \wedge R) = (P \vee Q) \wedge (P \vee R) \text{ is like } A \vee (B \wedge C) = (A \vee B) \wedge (A \vee C)$$

Translating between predicates and subsets

Let S be a set and $P(s), Q(s), R(s)$ be predicates of S

Let $A, B, C \subseteq S$ where $A = \{s \in S : P(s) \text{ is true}\}$

$B = \{s \in S : Q(s) \text{ is true}\}$

$C = \{s \in S : R(s) \text{ is true}\}$

For each $s \in S : P(s) \text{ is true} \Leftrightarrow s \in A$

$\rightarrow P(s) \wedge Q(s) \text{ is true} \Leftrightarrow P(s) \text{ is true and } Q(s) \text{ is true}$

$\Leftrightarrow s \in A \text{ and } s \in B$

$\Leftrightarrow s \in A \cap B$

$\rightarrow P(s) \vee Q(s) \text{ is true} \Leftrightarrow s \in A \cup B$

$\rightarrow P(s) \vee (Q(s) \wedge R(s)) \text{ is true} \Leftrightarrow s \in A \text{ or } s \in B \cap C$

$\Leftrightarrow s \in A \cup (B \cap C)$

$\rightarrow (P(s) \vee Q(s)) \wedge (P(s) \vee R(s)) \Leftrightarrow s \in (A \cup B) \cap (A \cup C)$

De Morgan's laws

To negate $P \wedge Q$, negate P and Q separately, and change $\wedge$ to $\vee$.

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e.g., $\neg(P \wedge Q) = (\neg P) \vee (\neg Q)$

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